University of Florida, Department of Mathematics

Topology First-Year Exam, January 2015, Part 1

Work the following problems and show all work. Support all statements to the best of your ability.

1. Show that there is no continuous function $f : [0,1] \to [0,1)$ which is onto.

2. Show that there is no continuous function $f : \mathbb{R} \to \{0,1\}$ which is onto.

3. Let X be a metric space and let $f : X \to X$ be a continuous function. Suppose that there is a point $x_0 \in X$ such that $f^n(x_0) \to z$ as $n \to \infty$. Show that $f(z) = z$.

4. Show that if $A \subset \mathbb{R}$ is connected, then A is an interval.

5. Prove that if X is compact Hausdorff, then X is a normal space.

6. State and prove the Contraction Mapping Theorem.

7. Suppose that X is a complete metric space that is countably infinite. Show that X has an isolated point.

Answer the following with complete definitions or statements or short proofs.

8. State the Intermediate Value Theorem.

9. State the Urysohn Lemma.

10. State the Tietze Extension Theorem.

11. State the Baire Category Theorem.

12. Give an example of a connected space X that is not path connected.

13. Let $n \geq 2$. Suppose that A is a countable subset of $\mathbb{R}^n$. Let $X = \mathbb{R}^n \setminus A$. Is X arcwise connected?

14. Suppose that $r > 0$ and that $B_r(x)$ is a ball of radius r centered at x in $\mathbb{R}^n$. Show that $B_r(x)$ is connected.

15. Show that $[0,1]$ is uncountable.